

CENTRAL BOARD OF SECONDARY EDUCATION**HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)****SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 1: EXPLORATION: ENTERING THE WORLD OF SECONDARY SCIENCE [SET C - NCERT EXEMPLAR SPECIAL]****TIME ALLOWED: 45 MINUTES****MAXIMUM MARKS: 25****PAPER CODE: 086/CH01/SET-C****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS (SET C - NCERT EXEMPLAR & CRITICAL THINKING):**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** NCERT Exemplar based Multiple Choice Questions with single correct option.
3. **Section B (Q16 – Q20):** Deep Assertion-Reasoning based questions testing scientific cause-and-effect.
4. **Section C (Q21 – Q25):** Case-Study / Laboratory Experimental Investigation questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** completely using a blue/black ballpoint pen.

SECTION A: NCERT EXEMPLAR MCQS (Q.1 TO Q.15)**[15 MARKS]**

Q.1 A student measures the period of oscillation T of a simple pendulum with lengths 20 cm, 40 cm, 60 cm, and 80 cm. A graph of T^2 versus pendulum length L yields:

- (A) A parabola curving upward
(B) A straight line passing through the origin
(C) A hyperbolic curve
(D) A horizontal line parallel to the length axis

Q.2 The mass of a solid block is measured as $m = (50.0 \pm 0.5)$ g and its volume is $V = (25.0 \pm 0.5) \text{ cm}^3$. What is the relative percentage error in its calculated density?

- (A) 1.0%
(B) 2.0%
(C) 3.0%
(D) 0.5%

Q.3 Which of the following measurements has exactly four significant figures?

- (A) 0.0025 kg
(B) 2.500 g
(C) 0.0250 L
(D) 2500 m

Q.4 When subtracting 1.2 cm from 15.345 cm, what should the result be recorded as, following rules of significant figures?

- (A) 14.145 cm
(B) 14.15 cm
(C) 14.1 cm
(D) 14 cm

Q.5 A micrometer screw gauge has a pitch of 0.5 mm and 50 divisions on its circular thimble. Its least count is:

- (A) 0.01 mm
(B) 0.05 mm
(C) 0.1 mm
(D) 0.001 mm

Q.6 Why is the metric SI system preferred over imperial units (inches, feet, pounds) in global scientific investigations?

- (A) It uses arbitrary conversion factors
(B) It is based on a coherent decimal base-10 system that standardizes measurements reproducibly across all nations
(C) It was invented earlier than any other system
(D) It only measures temperature

Q.7 A student accidentally spills dilute hydrochloric acid onto her hand. What is the immediate first-aid action to be taken in the laboratory?

- (A) Neutralize with concentrated sodium hydroxide
(B) Rinse immediately under a continuous stream of running tap water for several minutes
(C) Apply butter or oil
(D) Cover immediately with an airtight plastic bandage

Q.8 Which of the following physical quantities is dimensionless (has no physical units)?

- (A) Density
(B) Specific gravity (relative density)
(C) Velocity
(D) Acceleration

Q.9 A piece of cork of mass 6 g and density 0.25 g/cm^3 floats on water. What is the actual volume of the cork?

- (A) 1.5 cm^3
(B) 24 cm^3
(C) 12 cm^3
(D) 0.04 cm^3

Q.10 How does repeatable precision differ fundamentally from experimental accuracy?

- (A) Precision measures closeness to the true accepted standard, while accuracy measures agreement between repeated trials
(B) Accuracy measures closeness to the true accepted value, while precision measures the consistency and agreement among repeated measurements
(C) Accuracy and precision are identical physical terms
(D) Precision requires zero instruments

Q.11 If the slope of the T^2 versus L graph for a simple pendulum is calculated to be $4.02 \text{ s}^2/\text{m}$, what is the experimental value of g (take $\pi = 3.1416$)?

- (A) 9.82 m/s^2
(B) 10.2 m/s^2
(C) 9.50 m/s^2
(D) 8.90 m/s^2

Q.12 Which of the following is a fundamental base unit in the International System of Units (SI)?

- (A) Joule (J)
(B) Watt (W)
(C) Candela (cd)
(D) Pascal (Pa)

Q.13 A student measures the thickness of a single sheet of paper by measuring the total thickness of a 500-page ream as 5.0 cm. What is the thickness of one sheet?

- (A) 0.1 mm
(B) 0.01 mm
(C) 1.0 mm
(D) 0.05 mm

Q.14 What is the primary function of a wire gauze placed beneath a glass beaker on a tripod stand during heating?

- (A) To collect chemical spills
(B) To distribute heat evenly over the bottom of the beaker and prevent thermal shock breakage
(C) To act as a chemical catalyst
(D) To reduce boiling temperature

Q.15 Express the speed of light in vacuum ($300,000,000 \text{ m/s}$) in standard scientific notation:

- (A) $30 \times 10^7 \text{ m/s}$
(B) $3.0 \times 10^8 \text{ m/s}$
(C) $0.3 \times 10^9 \text{ m/s}$
(D) $3.0 \times 10^{-8} \text{ m/s}$

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).

(C) Assertion (A) is true but Reason (R) is false.

(D) Assertion (A) is false but Reason (R) is true (or both are false).

Q.16 Assertion (A): The least count of a measuring instrument represents the smallest measurement that can be accurately resolved by it.

Reason (R): A smaller least count generally leads to higher precision in measurement.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.17 Assertion (A): Random errors cannot be completely avoided even in the most carefully designed experiment.

Reason (R): Random errors arise from unpredictable microscopic variations in environmental conditions and minor observer judgment fluctuations.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.18 Assertion (A): A physical equation can be dimensionally correct yet physically incorrect.

Reason (R): Dimensionless constants (such as $\frac{1}{2}$, π) in an equation cannot be verified by dimensional analysis.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.19 Assertion (A): Flammable liquids like kerosene or alcohol can be extinguished using water.

Reason (R): Water has a high specific heat capacity that cools burning liquids.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

Q.20 Assertion (A): In a scientific controlled experiment, only one independent variable is altered at a time.

Reason (R): Altering multiple variables simultaneously makes it impossible to isolate which specific factor produced the observed change.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A). (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION C: EXPERIMENTAL INVESTIGATION & CASE STUDY (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL INVESTIGATION:

Case Study: Investigation of Thermal Expansion of Metals

A physics student investigates the linear thermal expansion of three metallic rods of identical initial length (1.000 m at 20°C): Rod A (Copper), Rod B (Aluminum), and Rod C (Iron). She places each rod into a steam jacket heated to 100°C (temperature increase $\Delta T = 80^\circ\text{C}$). The expansions measured using a dial gauge are:

- Rod A (Copper): 1.36 mm
- Rod B (Aluminum): 1.92 mm
- Rod C (Iron): 0.96 mm

Q.21 Which metal rod demonstrated the highest coefficient of linear expansion (α)?

- (A) Rod A (Copper) (B) Rod B (Aluminum)
(C) Rod C (Iron) (D) All three expanded equally

Q.22 What is the coefficient of linear expansion of Iron (Rod C) per degree Celsius?

- (A) $1.2 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ (B) $2.4 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$
(C) $1.7 \times 10^{-5} \text{ } ^\circ\text{C}^{-1}$ (D) $0.96 \times 10^{-3} \text{ } ^\circ\text{C}^{-1}$

Q.23 If a bimetallic strip made of Copper (Rod A) and Iron (Rod C) riveted together is heated, towards which metal will the strip bend?

- (A) Towards Copper, because Copper expands more (B) Towards Iron, because Iron expands less, forming the inner arc of curvature
(C) It remains completely straight (D) It twists into a corkscrew

Q.24 Why are gaps left between successive steel rails on railway tracks?

- (A) To save steel material cost (B) To allow thermal expansion during hot summer days without buckling the tracks
(C) To allow rainwater drainage (D) To reduce train speed

Q.25 Which practical device makes use of the differential thermal expansion of a bimetallic strip to regulate temperature?

- (A) Mercury barometer (B) Thermostat in electric irons and refrigerators
(C) Hydrometer (D) Bunsen burner air collar

CBSE CANDIDATE OMR ANSWER RESPONSE GRID (SET C)

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & EXEMPLAR EXPLANATIONS (SET C)

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.1	(B)	Since $T = 2\pi \sqrt{L/g}$, squaring both sides gives $T^2 = \left(\frac{4\pi^2}{g}\right) L$. This is of the form $y = mx$, representing a straight line passing through the origin.
Q.2	(C)	Relative error in density $\frac{\Delta \rho}{\rho} = \frac{\Delta m}{m} + \frac{\Delta V}{V} = \frac{0.5}{50.0} + \frac{0.5}{25.0} = 0.01 + 0.02 = 0.03 = 3.0\%$.
Q.3	(B)	Trailing zeros after a decimal point are significant. '2.500' has four significant figures (2, 5, 0, 0). Leading zeros in 0.0025 and 0.0250 are not significant.
Q.4	(C)	In addition and subtraction, the final result can have no more decimal places than the measurement with the fewest decimal places (1.2 cm has 1 decimal place; rounded to 14.1 cm).
Q.5	(A)	$\text{Least count} = \frac{\text{Pitch}}{\text{Number of circular divisions}} = \frac{0.5 \text{ mm}}{50} = 0.01 \text{ mm}$.
Q.6	(B)	The SI system is globally standardized, coherent, and uses decimal powers of ten, minimizing unit translation errors and simplifying scientific calculations.
Q.7	(B)	Flushing with large volumes of running water rapidly dilutes and washes away the acid. Never apply strong bases, as neutralization is highly exothermic and can cause thermal burns.
Q.8	(B)	Relative density is the ratio of density of a substance to density of water at 4°C. Being a ratio of two identical physical units, it is pure dimensionless.
Q.9	(B)	$V = \frac{\text{Mass}}{\text{Density}} = \frac{6 \text{ g}}{0.25 \text{ g/cm}^3} = 24 \text{ cm}^3$.
Q.10	(B)	Accuracy is how close a measured value is to the true or accepted reference value; precision is how close repeated measurements are to one another.
Q.11	(A)	$\text{Slope} = \frac{4\pi^2}{g} \Rightarrow g = \frac{4\pi^2}{\text{Slope}} = \frac{4(3.1416)^2}{4.02} = \frac{39.478}{4.02} \approx 9.82 \text{ m/s}^2$.
Q.12	(C)	Candela (cd) is the SI base unit of luminous intensity. Joule, Watt, and Pascal are derived units.
Q.13	(A)	Thickness per sheet $= \frac{5.0 \text{ cm}}{500} = 0.01 \text{ cm} = 0.1 \text{ mm}$.
Q.14	(B)	The wire gauze (often with a ceramic center) spreads the Bunsen flame heat uniformly, preventing localized hot spots that shatter glassware.
Q.15	(B)	$300,000,000 \text{ m/s} = 3.0 \times 10^8 \text{ m/s}$ in standard normalized scientific notation.
Q.16	(B)	Both statements are correct facts, but the reason states an advantage of small least count rather than explaining the operational definition of least count.
Q.17	(A)	Random errors follow a normal statistical distribution due to uncontrollable thermal, mechanical, or human limits, making them inevitable in empirical testing.
Q.18	(A)	For instance, $s = 2ut + t^2$ is dimensionally consistent ($[L] = [L]$) but physically wrong because the constant factor is erroneous.
Q.19	(D)	Assertion is completely false: Kerosene and alcohol are less dense than water; pouring water causes the burning liquid to float on top, spreading the fire! (Use sand or foam extinguisher).
Q.20	(A)	Fair testing requires holding all controlled variables fixed while systematically varying only the single factor under study.
Q.21	(B)	Since $\Delta L = L_0 \alpha \Delta T$, with identical initial length and temperature rise, Aluminum expanded the most (1.92 mm), indicating the highest α .
Q.22	(A)	$\alpha = \frac{\Delta L}{L_0 \Delta T} = \frac{0.00096 \text{ m}}{1.000 \text{ m} \times 80^\circ \text{C}} = 1.2 \times 10^{-5} \text{ } ^\circ \text{C}^{-1}$.

Q. No.	Correct	Scientific Rationale & NCERT Exemplar Analysis
Q.23	(B)	Copper expands more than Iron, forcing Copper to form the longer outer curve while Iron forms the shorter inner curve. Hence it bends towards Iron.
Q.24	(B)	Without expansion gaps, high summer temperatures cause thermal expansion stress that severely warps and buckles railway tracks, risking derailment.
Q.25	(B)	Thermostats use bending bimetallic strips to mechanically make and break electrical contacts as temperature fluctuates.